

The **AI Expense Tracker** project is built and ready for your BCA 5th Semester AI major project submission.

### What was created

A full-stack AI expense tracking app with:

| Module        | Purpose                                                             |
|---------------|---------------------------------------------------------------------|
| app.py        | Main entry with login/register, sidebar navigation, dark/light mode |
| database.py   | SQLAlchemy ORM – Users, Expenses, Bills, Predictions, Budgets       |
| auth.py       | Registration, login, logout with bcrypt hashing                     |
| expense.py    | Add, edit, delete, search, filter expenses                          |
| ocr.py        | EasyOCR bill scanner (store, date, amount, items)                   |
| prediction.py | ML pipeline – Linear Regression, Random Forest, Decision Tree       |
| charts.py     | Plotly charts (trends, pie, daily, weekly, comparison)              |
| budget.py     | Budget alerts at 80% / 100% with suggestions                        |
| utils.py      | Theming, currency formatting, AI insights                           |

**6 Streamlit pages:** Dashboard, Add Expense, OCR Scanner, Predictions, Analytics, Profile

### Generated assets:

- data/sample\_expenses.csv – 800 realistic expense records
- models/spending\_model.pkl – trained ML model
- database/expense.db – SQLite database (auto-created)

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### How to run

```
cd "d:\python_projects\AI Expense Tracker"
```

```
# Create virtual environment (recommended)
```

```
py -m venv venv
```

```
venv\Scripts\activate
```

```
# Install dependencies
```

```
pip install -r requirements.txt
```

```
# Generate dataset & train model (already done, but you can re-run)
```

```
py generate_dataset.py
```

```
py train_model.py
```

```
# Launch the app
```

```
streamlit run app.py
```

Open <http://localhost:8501>, register an account, set a budget in Profile, and start using the app.

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### Features for viva / presentation

1. **Authentication** – bcrypt password hashing, per-user data isolation
  2. **OCR (AI)** – EasyOCR extracts bill details from uploaded images
  3. **Machine Learning** – 3 models compared; best model saved with Joblib
  4. **Analytics** – 6 interactive Plotly charts
  5. **AI Insights** – month-over-month statistical analysis
  6. **Budget System** – warnings, critical alerts, category suggestions
  7. **Professional UI** – dark/light mode, cards, custom CSS, sidebar icons
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### Project report outline

The README includes sections for Introduction, Literature Survey, System Design, Implementation, Testing, Results, Conclusion, and Future Scope — suitable for your project report and viva.

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### Note on OCR

EasyOCR depends on PyTorch and is large (~2 GB). Core packages are installed; for OCR, ensure full install:

```
pip install easyocr torch torchvision opencv-python-headless
```

The first OCR scan may take a minute while EasyOCR downloads its language model.

If you want, I can add a demo user with sample expenses, a .gitignore, or a project report template next.

The app startup check succeeded. Streamlit launched at <http://localhost:8501> with no import or runtime errors. The non-zero exit code was from stopping the server after verification, not from an app failure.

To run it yourself:

```
cd "d:\python_projects\AI Expense Tracker"
```

```
streamlit run app.py
```

Register an account on first launch, then explore Dashboard, OCR Scanner, and Predictions.